

Objective

The objective of the Model Building phase is to train, evaluate, and compare multiple machine learning models for loan approval prediction. Different classification algorithms are implemented to identify the model that provides the highest prediction accuracy and reliability. The best-performing model is then saved for deployment in the Smart Lender web application.

Description

Model Building is a crucial stage in the machine learning workflow where preprocessed data is used to train predictive models. In this phase, several supervised machine learning algorithms are implemented and evaluated using appropriate performance metrics. The models are trained using the training dataset and tested on unseen data to assess their predictive capability.

The Model Building phase consists of the following stories:

Story 1: Decision Tree Model

A Decision Tree Classifier is developed to classify loan applications based on applicant information. The model is trained using the training dataset and evaluated on the testing dataset. Performance is measured using a confusion matrix, classification report, and accuracy score.

Story 2: Random Forest Model

A Random Forest Classifier is implemented by combining multiple decision trees to improve prediction accuracy and reduce overfitting. The model is trained and evaluated using the same dataset, and its performance is analyzed using evaluation metrics such as the confusion matrix, classification report, and accuracy score.

Story 3: K-Nearest Neighbors (KNN) Model

A K-Nearest Neighbors (KNN) Classifier is developed to classify loan applications based on the similarity between applicant records. Since KNN is a distance-based algorithm, it uses the scaled dataset for training and testing. Model performance is evaluated using standard classification metrics.

Story 4: Gradient Boosting (XGBoost) Model

A Gradient Boosting Classifier is implemented to improve prediction performance by combining multiple weak learners into a strong predictive model. The model is trained and evaluated on the prepared dataset, and its performance is compared with the other classification models.

Story 5: Evaluating Performance and Saving the Model

All trained models are compared using evaluation metrics such as accuracy score, confusion matrix, classification report, and 5-fold cross-validation. The model with the best overall performance is selected as the final prediction model. The selected model is saved as a **.pkl** file using the pickle library so that it can be integrated into the Flask web application for real-time loan approval prediction.

Machine Learning Models Used

- Decision Tree Classifier
 - Random Forest Classifier
 - K-Nearest Neighbors (KNN)
 - Gradient Boosting Classifier (used in place of XGBoost)
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Performance Evaluation Metrics

The following metrics are used to evaluate and compare the models:

- Accuracy Score
- Confusion Matrix
- Classification Report
- Precision
- Recall
- F1-Score

- 5-Fold Cross-Validation
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Tools and Libraries Used

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Pickle
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Expected Outcome

After completing the Model Building phase:

- Multiple machine learning models are successfully trained.
 - Each model is evaluated using appropriate performance metrics.
 - The performance of all models is compared.
 - The best-performing model is identified.
 - The selected model is saved as a **.pkl** file for deployment.
 - The saved model is ready for integration with the Flask web application.
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Conclusion

The Model Building phase successfully develops and evaluates multiple machine learning models for loan approval prediction. By comparing the performance of Decision Tree, Random Forest, K-Nearest Neighbors, and Gradient Boosting classifiers, the most accurate and reliable model is selected. The final model is saved as a serialized **.pkl** file, making it ready for deployment within the Smart Lender application to provide real-time loan approval predictions.